

Databases in the Era of Cloud Computing and Big Data

Part—1

In this series of two articles, we take a look at the directions in which databases are evolving, driven by the twin factors of the 'Cloud' and 'Big Data'. In this first part, let's start with a quick look at cloud computing, and discuss the 'Big Data' explosion, focusing on its impact on database systems. I will trace the evolution of databases from simple flat-file to enterprise RDBMS and NoSQL databases. I will briefly touch on the NoSQL movement, CAP theorem and ACID vs BASE semantics. The second part will uncover the challenges for next-generation databases, and see how database research aims to address these needs, using emerging technologies like non-volatile memory and the many-core revolution.

If asked, "What are the two mega-trends in the computing industry likely to be in the coming decade?" most IT industry folks would name 'Big Data' and 'Cloud Computing' as the driving forces shaping the industry today. The computing world is shifting from enterprise-centric to data-centric workloads driven by the 'Big Data' revolution, while cloud computing is becoming mainstream, reinventing utility/elastic computing as the new mantra for the IT industry. The influence of these two

paradigms is driving the revolution in different fields of the computing industry, such as processor/platform architectures, programming models, programming languages and software stacks, to name just a few. After all, no one would have paid attention to 'Map-Reduce' as a programming paradigm, if it was not for Web 2.0 and the shift towards cloud computing.

The field of databases is no exception, and has been influenced heavily by these two driving forces. Cloud computing is driving

